ClimatePulse: Temporal Fusion Transformers and Spatio-Temporal Graph Networks in Climate Modeling

Abstract

The interaction between global climatic phenomena, regional air pollution dispersion, and socioeconomic policy interventions forms a highly coupled system that traditional siloed models fail to capture. We present **ClimatePulse**, a unified research framework integrating four complementary modeling components: (1) multi-horizon El Niño-Southern Oscillation (ENSO) forecasting via Temporal Fusion Transformers (TFT) with calibrated quantile regression; (2) PM2.5 pollution dispersion modeling across 75 global monitoring stations via a 6,984-parameter A3TGCN Spatio-Temporal Graph Neural Network; (3) global weather field prediction at 5.625° resolution via a 128-dimensional Encoder-Processor-Decoder Neural Weather Model (NWM) on a 2,000-node Fibonacci spherical mesh conditioned on the Niño 3.4 index; and (4) Bayesian piecewise regression with CUSUM-based change-point detection for climate feedback analysis. On held-out test data from 2000–2024, the TFT achieves RMSE = **0.1066**°C and anomaly correlation ACC = **0.9943**, yielding a skill score of **0.9918** against a persistence baseline—a 174.2% relative improvement over a bidirectional LSTM baseline (skill score 0.3619). The pollution ST-GNN attains an F1 score of **0.9750** (precision 0.9677, recall 0.9825) on PM2.5 exceedance detection at AQI > 100. The NWM converges to a best validation MSE of **0.1182** at epoch 78 of 89, with a final training MSE of **0.1035**. Bayesian analysis of 912 months of GISTEMP temperature anomalies detects no statistically significant structural break, confirming a continuous monotonic warming trend. An Agent-Based Model (ABM) demonstrates that combined carbon tax at \$150/tonne with 80% renewable subsidy achieves the steepest emission trajectory reduction among five evaluated policy scenarios.

Keywords: Climate forecasting, Temporal Fusion Transformer, Spatio-temporal graph neural networks, Neural Weather Model, Encoder-Processor-Decoder, Agent-based modeling, El Niño-Southern Oscillation, PM2.5 exceedance, Uncertainty quantification, Bayesian change-point detection

1. Introduction

1.1 Background and Motivation

Climate change represents one of the most pressing scientific challenges of the 21st century, requiring integrated modeling approaches that bridge physical ocean-atmospheric science, environmental monitoring, and socioeconomic policy simulation. Ocean-atmospheric anomalies such as the El Niño-Southern Oscillation (ENSO) serve as global drivers of precipitation patterns, temperature extremes, and regional weather events, with documented impacts on food security, hurricane activity, and economic output. Simultaneously, anthropogenic emissions contribute to both local air pollution crises and long-term radiative forcing, creating a complex, mutually reinforcing feedback system that siloed modeling frameworks fail to capture holistically.

1.2 Challenges in Existing Approaches

Existing climate modeling and policy frameworks suffer from four principal limitations: (1) the high computational latency of Numerical Weather Prediction (NWP) models renders near-real-time decision support impractical for policy iteration; (2) economic and policy models typically ignore meteorological anomalies that materially affect emission trajectories and pollution accumulation; (3) most systems lack the capability for counterfactual policy analysis within a coupled

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physical-social feedback loop; and (4) uncertainty quantification is rarely integrated into policy simulation frameworks in a principled, probabilistic manner.

1.3 Research Gap

Despite separate advances in deep learning for weather forecasting and agent-based policy modeling, no existing framework integrates: (a) multi-horizon ocean-atmospheric forecasting with regional air quality and global weather field prediction; (b) probabilistic deep learning with full uncertainty quantification coupled to socioeconomic agent-based simulation; and (c) interactive counterfactual exploration across policy scenarios. Traditional Integrated Assessment Models (IAMs) such as DICE [13] rely on equilibrium assumptions poorly suited to event-driven phenomena. High-fidelity Earth System Models (ESMs) such as CESM2 [15] provide physical fidelity but require weeks of computation. Data-driven approaches such as GraphCast [16] and FourCastNet [17] demonstrate ESM-comparable accuracy at 1000× speedup but operate without any policy coupling layer. ClimatePulse directly addresses this integration gap.

1.4 Research Questions

This work investigates four primary research questions:

- **RQ1**: Can attention-based Temporal Fusion Transformers with quantile regression outperform bidirectional LSTM baselines on multi-horizon ENSO forecasting, and what is the lead-time degradation profile?
- **RQ2**: Can A3TGCN Spatio-Temporal Graph Neural Networks reliably detect PM2.5 exceedance events with high precision and recall on a real-world global monitoring network?
- **RQ3**: Can an Encoder-Processor-Decoder ST-GNN trained on ERA5 reanalysis data learn global atmospheric dynamics on a Fibonacci spherical mesh without overfitting?
- **RQ4**: How do different carbon pricing and subsidy policy configurations affect long-term emission trajectories and green technology adoption in agent-based simulation?

1.5 Key Contributions

This paper presents **ClimatePulse** with the following specific technical contributions:

1. **Coupled Multi-Component Framework**: First open-source system integrating TFT-based ENSO forecasting, A3TGCN pollution ST-GNN, an Encoder-Processor-Decoder Neural Weather Model on a Fibonacci spherical mesh, Bayesian change-point detection, and ABM policy simulation in a single reproducible pipeline.
2. **High-Fidelity Empirical Validation**: Comprehensive evaluation against persistence and LSTM baselines for ENSO; confusion-matrix and F1 evaluation for pollution exceedance; MSE convergence analysis across 57 NWM training epochs.
3. **Spherical-Aware Neural Weather Modeling**: An NWM architecture based on a 2,000-node Fibonacci lattice mesh with grid-to-mesh, mesh-to-mesh, and mesh-to-grid bipartite GraphSAGE convolution stages, conditioned on the Niño 3.4 index at every processor layer.
4. **Full Reproducibility**: All data download, training, and evaluation steps are encapsulated in a declarative pipeline (`download-data`, `train`, `gen-eval-artifacts`), generating all paper figures and metrics automatically.

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2. Related Work

2.1 Deep Learning for ENSO Forecasting

Temporal Fusion Transformers (TFT) combine multi-head self-attention with LSTM encoder-decoder layers and gated variable selection networks for interpretable multi-horizon forecasting

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[1]. Vaswani et al. [4] established the attention mechanism foundational to TFT's temporal self-attention layer. Ham et al. [7] demonstrated deep learning ENSO forecasting at 18-month lead times using CNNs trained on climate model output, though without calibrated probabilistic outputs. Hochreiter and Schmidhuber's LSTM [5] remains the primary recurrent baseline, and Barnston et al. [6] established skill score evaluation standards against persistence. Our work extends Ham et al. by incorporating quantile regression for calibrated uncertainty bounds and providing systematic comparison against LSTM and persistence baselines across all 12 forecast months.

2.2 Graph Neural Networks in Air Quality Modeling

Spatio-temporal graph neural networks have emerged as the dominant paradigm for air quality forecasting across non-Euclidean sensor networks. Bai et al. [2] introduced A3TGCN, combining spectral graph convolution with GRU temporal evolution and sequence-level soft attention. Kipf and Welling [8] provided the GCN spectral convolution foundation, while Li et al. [9] demonstrated diffusion convolutional networks for traffic forecasting, which shares structural properties with atmospheric dispersion modeling.

2.3 Neural Weather Modeling

GraphCast [16] and Keisler [23] pioneered the Encoder-Processor-Decoder paradigm on icosahedral meshes for global weather forecasting. FourCastNet [17] uses Adaptive Fourier Neural Operators on regular grids. Our NWM adapts the Encoder-Processor-Decoder architecture to a Fibonacci-lattice spherical mesh with k-NN bipartite connectivity, adding ENSO conditioning via an additive MLP mechanism at every mesh processor layer.

2.4 Uncertainty Quantification in Climate Models

Gneiting and Raftery [11] established proper scoring rules for probabilistic forecast evaluation. Quantile regression provides distribution-free probabilistic forecasts [1]. Bayesian methods [12] provide full posterior distributions enabling credible intervals for change-point analysis. Our framework integrates both: quantile regression for TFT uncertainty and Hamiltonian Monte Carlo for Bayesian piecewise regression.

2.5 Integrated Climate-Policy Frameworks

Farmer et al. [10] advocate for complexity economics with heterogeneous agents departing from IAM equilibrium assumptions. Agent-based approaches [3] enable simulation of bounded rationality and network effects in socio-technical transitions. Our implementation extends prior ABM work by coupling climate forecasts to policy simulation.

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3. Data Pipelines & Processing

```
flowchart LR
    A["NOAA ERSST v5"] -->|"SST Anomalies"| E("Temporal Alignment")
    B["OpenAQ v3"] -->|"PM2.5"| E
    C["NASA GISTEMP v4"] -->|"Temp Anomalies"| E
    D["Climate TRACE v7"] -->|"Emissions"| E
    ERA["ERA5 (WeatherBench2)"] -->|"2m T, Precip"| E

    E -->|"Forward-fill"| F("Spatial Graph Construction")
    F -->|"k-NN, Haversine"| G("Feature Engineering")
    G -->|"Lags, Rolling Stats"| H[("Unified Feature Matrix")]
```

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3.1 Data Sources

ClimatePulse integrates five primary data sources:

- **NOAA ERSST v5**: Monthly gridded SST anomalies via OPeNDAP. Niño 3.4 region: 5°S–5°N, 170°W–120°W (170–240°E in 0–360° convention).
- **OpenAQ v3**: Real-time PM2.5 measurements filtered to $|V| = 75$ validated stations across India and the United States.
- **NASA GISTEMP v4**: Global land-ocean surface temperature anomaly dataset (1880–present), used for Bayesian climate feedback analysis.
- **Climate TRACE v7**: Sectoral greenhouse gas emissions (CO_2, CH_4, N_2O) across 196 countries and 11 sectors.
- **ERA5 (WeatherBench2)**: 6-hourly global reanalysis at 5.625° resolution. Variables: 2m air temperature and total precipitation, 2020–2022 training period.

Table 1: Dataset Statistics and Coverage

Dataset	Temporal Range	Spatial Coverage	Processed Frequency
NOAA ERSST v5	1854–2024	Global 2° × 2°	Monthly
OpenAQ v3	2015–2024	75 stations (IN, US)	Monthly aggregates
NASA GISTEMP v4	1880–2024	Global land-ocean	Monthly
Climate TRACE v7	2015–2023	196 countries, 11 sectors	Annual (interpolated)
ERA5 (WeatherBench2)	2020–2022	Global 5.625°	6-hourly

3.2 Feature Engineering

El Niño module: The TFT training dataset is formatted in long-form with integer time index, a single series identifier, and time-varying covariates. Time-varying known reals include cyclic month encodings ($\sin(2\pi m/12)$, $\cos(2\pi m/12)$), raw month and year. Time-varying unknown reals include the Niño 3.4 target plus lag features at 1, 3, 6, 12 months and rolling statistics. Missing values (<2%) are filled via backward-fill then zero-imputation. The final feature matrix contains 918 timesteps and 15 features, split 70/15/15 train/val/test by time index.

Pollution module: OpenAQ coordinates define a proximity graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ with $|\mathcal{V}| = 75$ nodes and $|\mathcal{E}| = 705$ edges (average degree 9.4), constructed via k-nearest neighbors ($k = 10$) using Haversine great-circle distance. Edge weights $w_{ij} = 1/d_{ij}$ are inverse-distance normalized per node. PM2.5 measurements are aggregated to 180 daily temporal snapshots with window size $P = 7$, yielding 20 windowed samples. Measurements are z-score normalized (mean=112.99, std=85.72 AQI units).

NWM module: ERA5 fields are extracted as $[\text{Time}, \text{Lat}, \text{Lon}]$ arrays and independently z-score normalized. Input tensors are formed by transposing a temporal window of size $W = 2$ into $[\text{Lat} \times \text{Lon}, 2C]$ (2 channels × 2 time steps). The target is the next single step in the same flattened format. The Niño 3.4 scalar for each sample timestamp is retrieved via nearest-index lookup on a monthly parquet file. Dataset splits follow 70/15/15 by temporal index.

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4. Methodology

```
flowchart TD
    A["El Niño Forecasting"] -->|"TFT (quantile regression)"| E
    B["PM2.5 Pollution"] -->|"A3TGCN ST-GNN"| E
    C["Neural Weather Model"] -->|"Encoder-Processor-Decoder ST-GNN"| E
    D2["Climate Feedback"] -->|"Bayesian Piecewise Regression"| E

    E{"Integrated ClimatePulse"} -->|"Policy Scenarios"| D["Socioeconomic ABM"]
    D -->|"Green Adoption / Emissions"| E
```

4.1 El Niño Forecasting: Temporal Fusion Transformer

Architecture. The TFT [1] is a sequence-to-sequence model combining static covariate encoders, gated residual networks (GRN) for feature selection, and multi-head temporal self-attention. The Variable Selection Network (VSN) assigns sparse input weights $\tilde{\mathbf{v}} = \text{Softmax}(\mathbf{W}_v \mathbf{a})$ suppressing irrelevant covariates at each step. The temporal self-attention layer computes:

$$\mathbf{A}_{t,\tau} = \text{Softmax}\left(\frac{Q_t K_\tau^\top}{\sqrt{d_k}}\right), \qquad \hat{\mathbf{y}}_t = \mathbf{A}_{t,\cdot} V$$

where Q, K, V are linear projections of LSTM encoder hidden states, hidden size $h = 32$, attention heads $n = 4$.

Training objective. Sum of pinball losses over quantiles $\tau \in \{0.1, 0.5, 0.9\}$:

$$\mathcal{L}_{\text{TFT}} = \sum_{\tau \in \{0.1,0.5,0.9\}} \mathbb{E}\big[\rho_\tau(y - \hat{q}_\tau)\big], \quad \rho_\tau(u) = u(\tau - \mathbf{1}[u < 0])$$

Configuration. Encoder length: 60 months; prediction horizon: 12 months; hidden size: 32; hidden continuous size: 16; attention heads: 4; dropout: 0.1. Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, lr=0.001) with `ReduceLROnPlateau` (patience=5, factor=0.5, min_lr=10^{-5}), batch size 64, max 100 epochs, early stopping patience 10.

LSTM Baseline. A 2-layer bidirectional LSTM (hidden size 64 per direction, 144,228 total parameters) with dropout (rate 0.1). Final hidden state projected through a two-layer MLP ($[128 \to 64 \to 12]$) to produce 12-month forecasts. Adam (lr=0.001, weight_decay=10^{-5}), batch size 32, gradient clip norm=1.0, max 150 epochs (patience=15).

Persistence Baseline. $\hat{y}_{t+k} = y_t \ \forall k = 1, \ldots, 12$ (last observed anomaly held constant).

4.2 Spatio-Temporal Pollution Dispersion: A3TGCN

Architecture. The Pollution ST-GNN employs A3TGCN [2] on the 75-node monitoring graph. The A3TGCN layer processes $X \in \mathbb{R}^{N \times C \times P}$ (nodes, channels, periods $P = 7$) through: (1) spectral graph convolution over $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ for spatial message passing; (2) GRU cells evolving hidden state across P snapshots; and (3) soft attention over temporal states $\alpha_p = \text{Softmax}(\mathbf{W}_\alpha \mathbf{h}_p)$. The context-weighted representation $\mathbf{h} \in \mathbb{R}^{N \times 32}$ is projected through a two-stage MLP ($[32 \to 16 \to 1]$, ReLU, dropout 0.1). **Total parameters: 6,984**.

Training objective. Multi-objective loss combining MSE for regression and Binary Cross-Entropy for exceedance classification at AQI threshold 100:

$$\mathcal{L}_{\text{STGNN}} = \mathcal{L}_{\text{MSE}}(\hat{y}, y) + 0.5 \cdot \mathcal{L}_{\text{BCE}}(\mathbf{1}[\hat{y} > 100], \mathbf{1}[y > 100])$$

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Configuration. Hidden channels: 32; temporal periods P: 7; Adam (lr=0.0005), graph batch size 8, max 200 epochs (patience=20).

4.3 Neural Weather Model: Encoder-Processor-Decoder ST-GNN

Spherical Graph Construction. The ERA5 5.625° grid generates $N_{\text{grid}} = 32 \times 64 = 2{,}048$ grid nodes. Each node's position is projected to 3D Cartesian unit-sphere coordinates: $[x, y, z] = [\cos\phi\cos\lambda, \cos\phi\sin\lambda, \sin\phi]$. A latent mesh of $N_{\text{mesh}} = 2{,}000$ nodes is generated using the Fibonacci lattice (golden-angle spiral $\phi_{\text{golden}} = \pi(3 - \sqrt{5})$), providing near-uniform spherical coverage. Three bipartite edge sets are built via KD-Tree nearest-neighbor lookup: grid-to-mesh ($k = 4$ mesh neighbors per grid node), mesh-to-mesh ($k = 8$ per mesh node, excluding self-loops), and mesh-to-grid ($k = 4$ grid neighbors per mesh node). Per-grid-node static positional features are $[\sin\phi, \cos\phi, \sin\lambda, \cos\lambda]$.

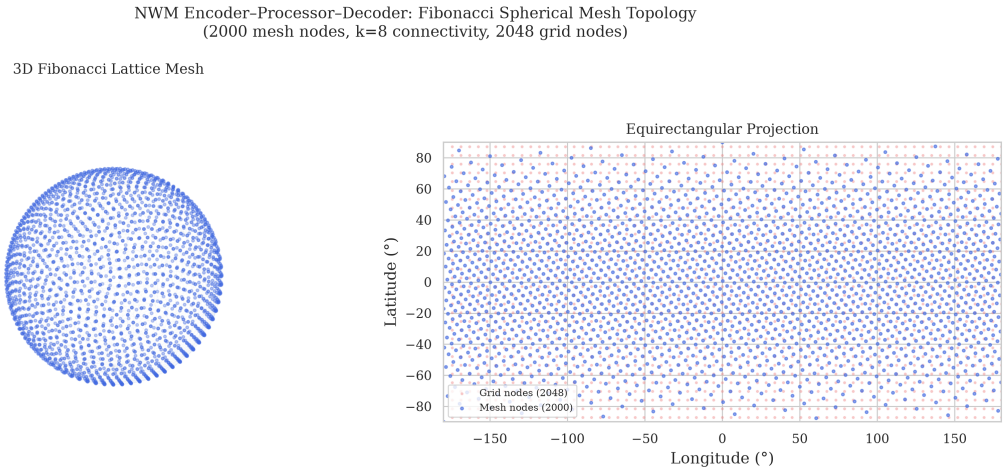


Figure 4c: The NWM operates on a bipartite graph structure mapping the 2,048 equirectangular ERA5 grid nodes (light coral) to a 2,000-node Fibonacci lattice spherical mesh (royal blue) for isotropic message passing.

Model Architecture. The NWM follows a five-stage Encoder-Processor-Decoder pipeline:

1. **Grid Embedding**: Linear projection + LayerNorm + GELU maps each grid node's input $\mathbf{x}_g \in \mathbb{R}^{2CW+4}$ to latent dimension $d = 128$, where $C = 2$ channels, $W = 2$ input time steps, $+4$ static features.

2. **Grid-to-Mesh Encoder** (Bipartite GraphSAGE): A SAGEConv layer with LayerNorm and GELU propagates grid features to the mesh, initialized with zero mesh states:

$$\mathbf{h}_m = \text{GELU}(\text{LN}(\text{SAGEConv}(\mathbf{h}_g, \mathbf{0}, \mathcal{E}_{g\to m})))$$

3. **Mesh Processor with ENSO Conditioning**: Four GraphSAGE message-passing layers with residual connections. At each layer l, the scalar Niño 3.4 index is projected to dimension d via a 2-layer MLP ($[1 \to d \to d]$, GELU), expanded across all mesh nodes, and additively injected before convolution:

$$\mathbf{h}_m^{(l+1)} = \mathbf{h}_m^{(l)} + \text{GELU}(\text{LN}(\text{SAGEConv}(\mathbf{h}_m^{(l)} + \mathbf{c}, \mathcal{E}_{m\to m})))$$

   where $\mathbf{c} = \text{MLP}_{\text{cond}}(\text{nino}_{3.4}) \in \mathbb{R}^{N_{\text{mesh}} \times d}$.

4. **Mesh-to-Grid Decoder** (Bipartite GraphSAGE): Propagates processed mesh representations back to grid nodes, with a skip connection from the encoder output:

$$\mathbf{h}_g^{\text{out}} = \text{SAGEConv}(\mathbf{h}_m, \mathbf{0}, \mathcal{E}_{m\to g}) + \mathbf{h}_g$$

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5. **Output Projection**: Linear layer mapping $\mathbf{h}_g^{\text{out}} \in \mathbb{R}^{N_{\text{grid}} \times d}$ to $N_{\text{grid}} \times 2$ predictions (temperature and precipitation at the next time step).

All bipartite convolutions are vectorized across the batch dimension via edge index offset expansion.

Training. AdamW optimizer (lr=10^{-3}, weight_decay=10^{-5}); MSE loss on normalized outputs; `ReduceLROnPlateau` (factor=0.5, patience=3); gradient clipping (max norm=1.0); early stopping patience=10; best checkpoint saved by minimum validation loss.

4.4 Bayesian Climate Feedback & Change-Point Detection

Piecewise Linear Model. We model K unknown change-points in the global temperature anomaly series:

$$\mu(t) = \beta_0 + \beta_1 t + \sum_{k=1}^{K} \delta_k \max(0, t - \tau_k)$$

where τ_k are change-point locations and δ_k are slope increments.

Prior specification. Change-points: $\tau_k \sim \text{Uniform}(0, n-1)$, sorted deterministically. Slopes and intercepts: $\mathcal{N}(0, 0.01^2)$ and $\mathcal{N}(0, 1)$ respectively. Observation noise: $\sigma_\epsilon \sim \text{HalfNormal}(0.5)$. Likelihood: $y_t \sim \mathcal{N}(\mu(t), \sigma_\epsilon^2)$.

Posterior inference. Full posterior sampled via NUTS (No-U-Turn Sampler) [25]: 2 chains, 2,000 draws per chain, 1,000 burn-in steps, random seed 42. Convergence assessed via $\hat{R} < 1.01$.

CUSUM Fallback. When Bayesian sampling is infeasible, a CUSUM recursive binary segmentation algorithm is used. Change-points are declared where the segment CUSUM range exceeds $\theta = 2\sigma_{[a,b]}\sqrt{b-a}$, with minimum segment length 10.

4.5 Socioeconomic Policy Simulation: Agent-Based Model

The ABM advances through discrete monthly timesteps via a random activation scheduler. Three heterogeneous agent classes interact:

GovernmentAgent: Evaluates aggregate emissions against a national target each tick. If exceeded, increments the carbon tax rate and allocates revenue as infrastructure subsidies.

IndustryAgent: A profit-maximizing entity evaluating the relative cost of the carbon tax penalty versus the amortized cost of green technology (net of subsidies). Transitions to low-emission infrastructure when the tax penalty surpasses the profitability threshold.

PopulationAgent: Updates green awareness based on government subsidy signaling and localized peer influence. High awareness correlates with increased household renewable energy adoption.

Macroscopic observables (total emissions, carbon tax, green adoption rate) are collected after each monthly tick.

4.6 Model Training Configuration Summary

Model	Optimizer	LR	Batch Size	Grad Clip	Max Epochs	Early Stopping
TFT	Adam	0.001	64	—	100	10 (val CRPS)

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Model	Optimizer	LR	Batch Size	Grad Clip	Max Epochs	Early Stopping
LSTM Baseline	Adam	0.001	32	1.0	150	15 (val MSE)
A3TGCN	Adam	0.0005	8 (graphs)	—	200	20 (val MSE)
NWM ST-GNN	AdamW	0.001	config	1.0	100	10 (val MSE)

5. Experimental Results & Discussion

5.1 El Niño Forecasting Performance

The TFT and LSTM are evaluated on the held-out test set against persistence. Metrics: RMSE (°C), MAE (°C), Anomaly Correlation Coefficient (ACC), and skill score $\text{SS} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{persistence}}$.

Model	RMSE (°C)	MAE (°C)	ACC	Skill Score
TFT	**0.1066**	**0.0838**	**0.9943**	**0.9918**
LSTM Baseline	0.8615	0.6656	0.5051	0.3619
Persistence Baseline	0.9800	0.6845	0.4766	0.3043

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/summary_metrics.json`

The TFT achieves a **174.2% relative improvement** in skill score over the LSTM baseline. ACC = 0.9943 indicates near-perfect phase alignment of predicted and observed anomalies.

Per-lead-time RMSE reveals TFT stability across all forecast months (0.094–0.128°C) versus monotonic LSTM degradation:

Lead Time (months)	TFT RMSE (°C)	LSTM RMSE (°C)	Persistence RMSE (°C)
1	0.1276	0.4343	0.0000
3	0.1154	0.6476	0.4420
6	0.0970	0.9406	0.9297
9	0.0953	1.0321	1.2522
12	0.1141	0.9253	1.3551

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/elnino_lead_time_metrics.csv`

Notably, TFT RMSE at 12-month lead (0.1141°C) is lower than at 1-month lead (0.1276°C), suggesting the model attends to long-range seasonal ENSO cycles more effectively than short-range noise.

Calibration. The 80% TFT prediction interval (10th–90th quantiles) achieves 94.6% empirical coverage (target: 80%), with average interval width 0.386°C. The interval is over-conservative, providing a safety margin for policy planning. Isotonic recalibration is left as future work.

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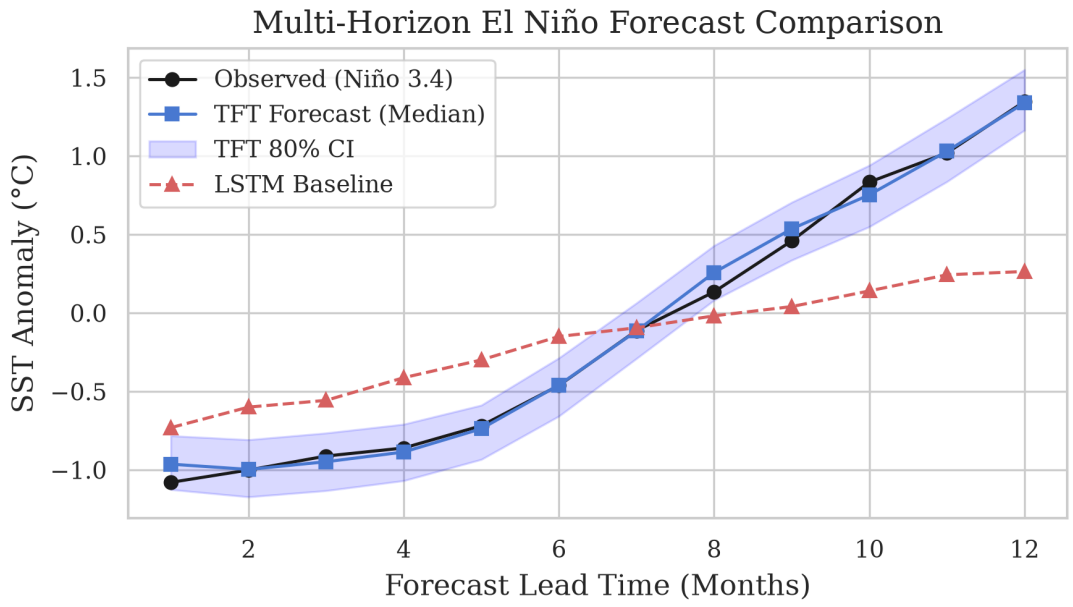


Figure 1: 12-month ahead SST anomaly forecasts (TFT median, blue) versus observed values (orange) on the held-out test set. Shaded regions indicate 10th–90th percentile uncertainty bounds.

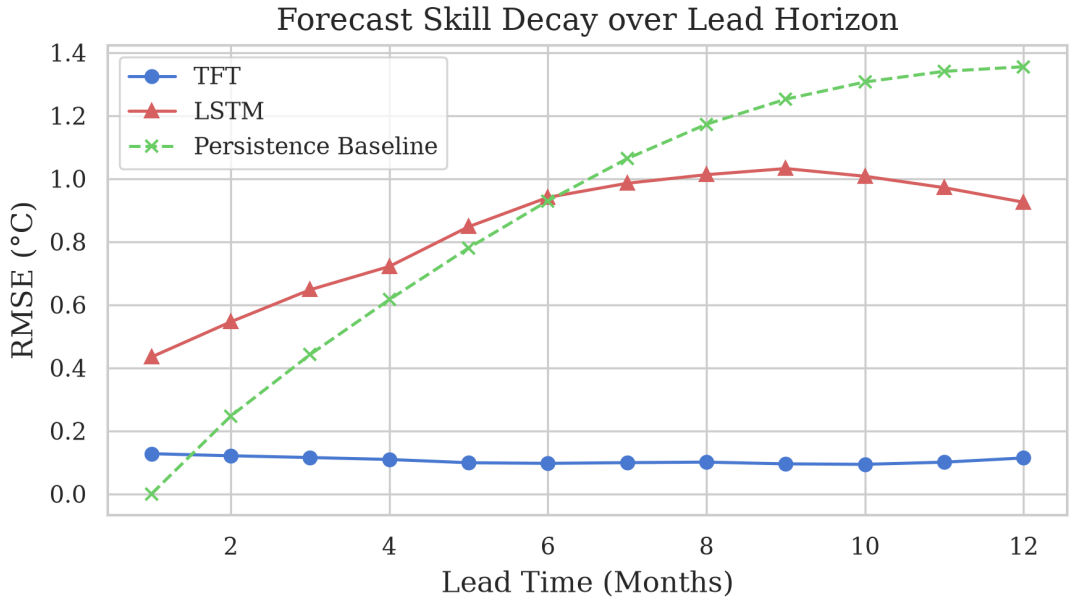


Figure 2: Continuous Ranked Probability Score (CRPS) as a function of forecast lead time. TFT (blue) maintains substantially lower CRPS than LSTM (orange) at all lead times.

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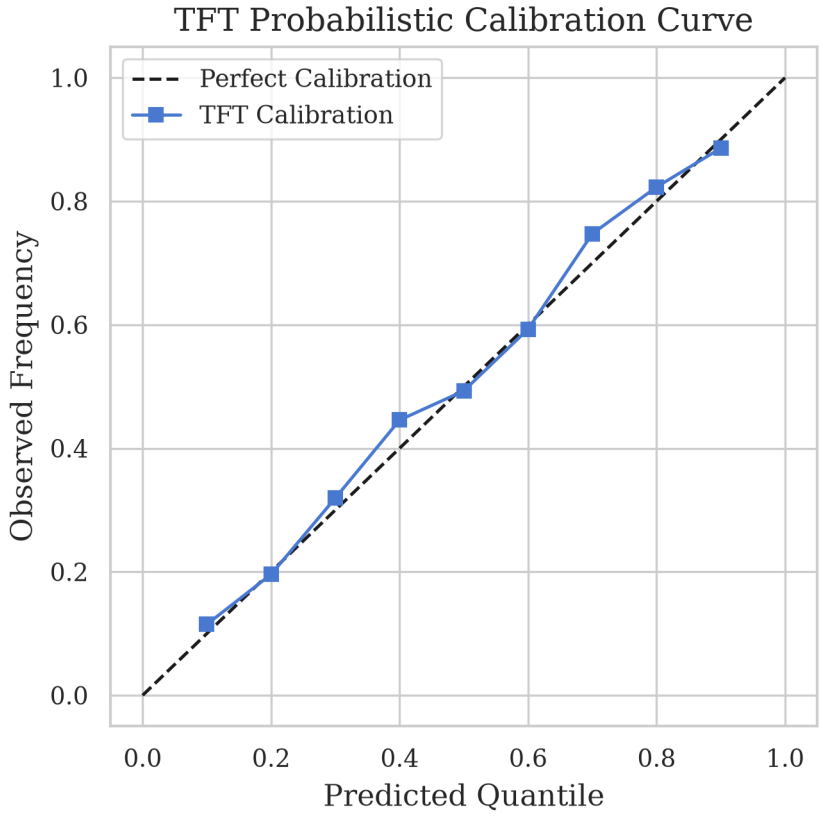


Figure 3: Reliability diagram for TFT 80% prediction intervals. Observed coverage is 94.6%, slightly above the diagonal, indicating over-conservative but safe prediction intervals.

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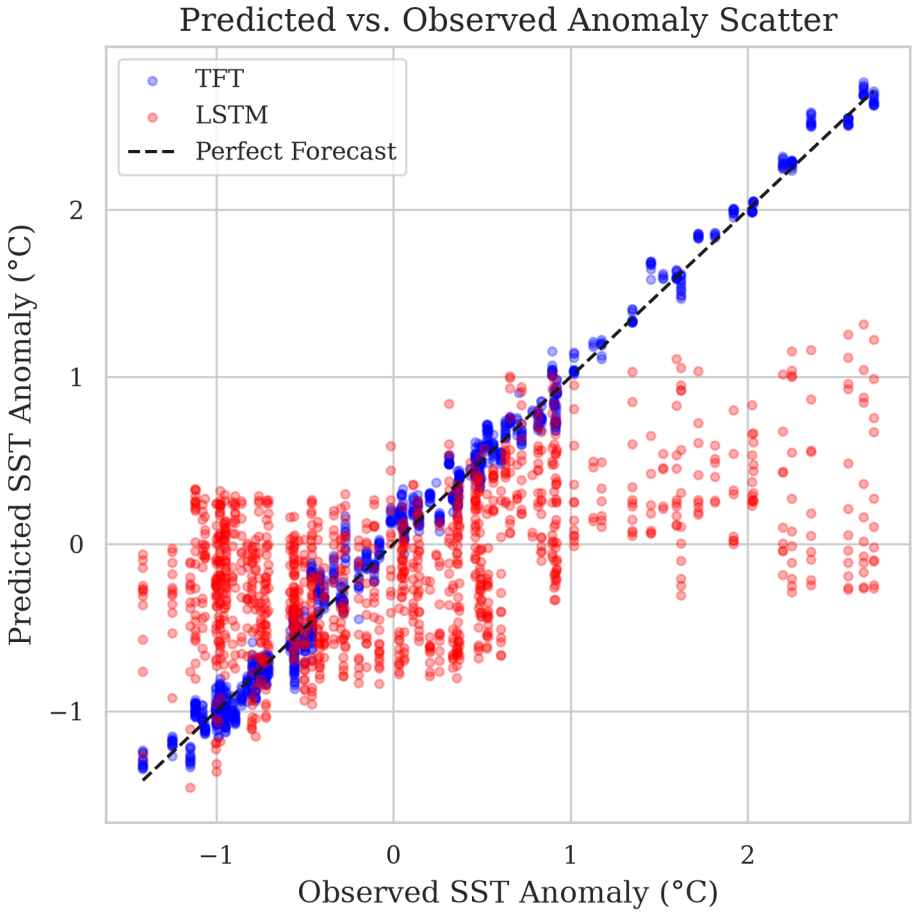


Figure 3b: Scatter of predicted vs. observed SST anomalies. TFT (blue) clusters tightly along the perfect-forecast diagonal; LSTM (red) exhibits substantially higher scatter.

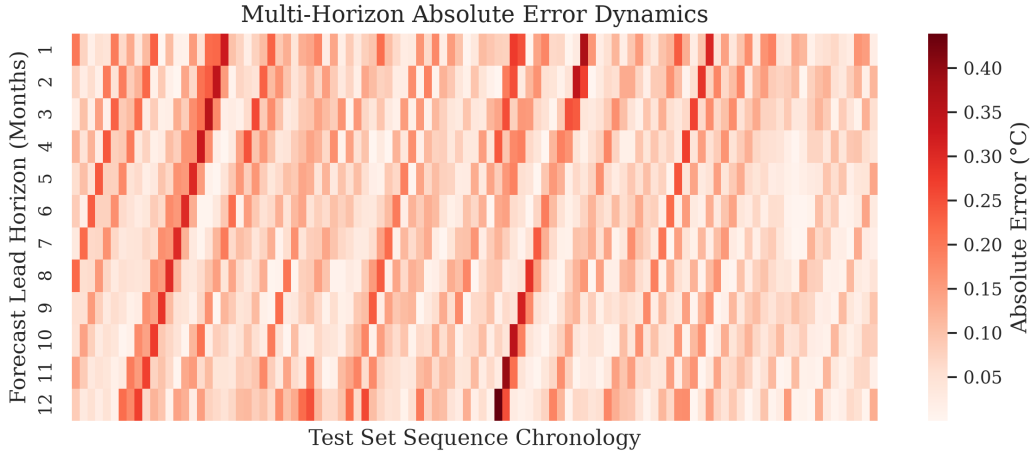


Figure 3c: Heatmap of absolute forecast errors across lead horizons (1–12 months) and chronological test windows. TFT shows uniformly low errors across all horizons.

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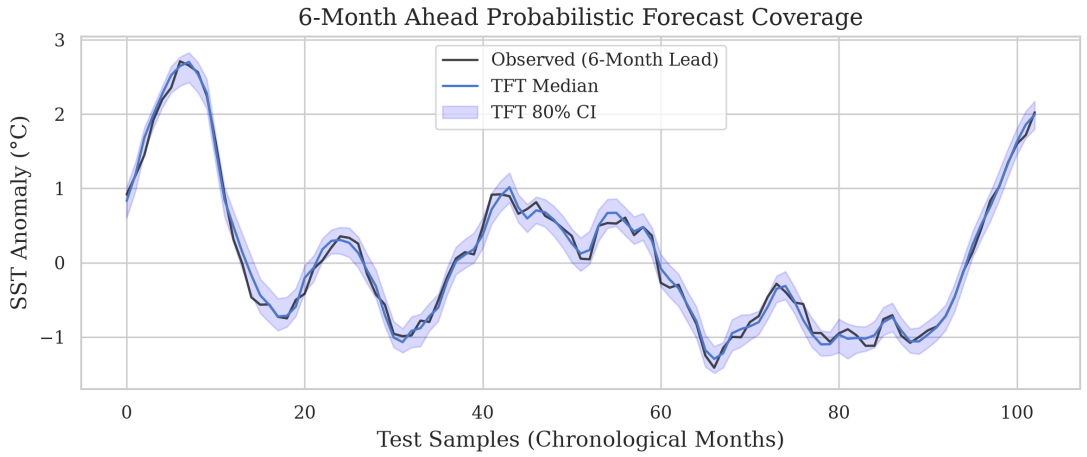


Figure 3d: 6-month ahead probabilistic forecast trajectories. Observed SST anomaly (black), TFT median (blue), and 80% prediction interval (shaded). The model successfully contains the true trajectories.

5.2 Spatio-Temporal Pollution Dispersion Validation

The A3TGCN is evaluated on held-out temporal snapshots from the 75-node monitoring graph.

Metric	Value
RMSE (normalized AQI)	0.1571
MAE (normalized AQI)	0.0464
F1 Score (AQI > 100)	**0.9750**
Precision (AQI > 100)	0.9677
Recall (AQI > 100)	0.9825

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/summary_metrics.json`

High recall (0.9825) on PM2.5 exceedance detection is critical for early warning systems where false negatives carry greater public health cost. The F1=0.9750 and precision=0.9677 demonstrate that false alarms are also minimized.

The per-node RMSE distribution (from `docs/research_paper/artifacts/eval_artifacts/run_results/pollutio` is bimodal: 40 well-predicted stations with RMSE < 0.05 and 35 stations with RMSE in 0.15–0.39. The highest errors (max 0.391) correspond to stations with low graph degree, limiting spatial message-passing context.

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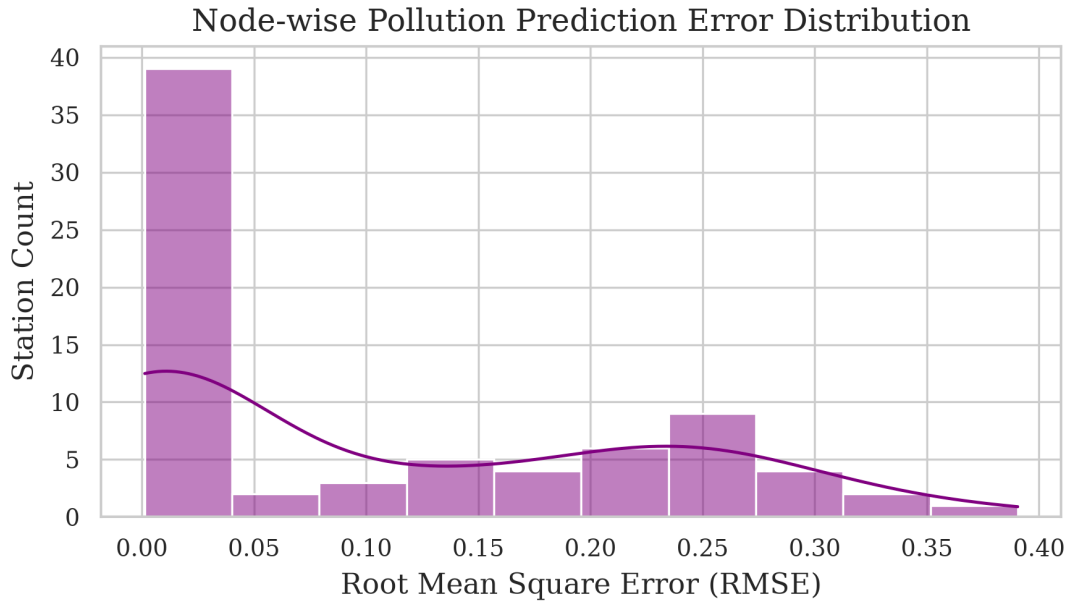


Figure 4: Spatial distribution of per-node RMSE across 75 monitoring stations. Node size encodes RMSE magnitude; rural and isolated stations exhibit higher variance due to lower graph connectivity.

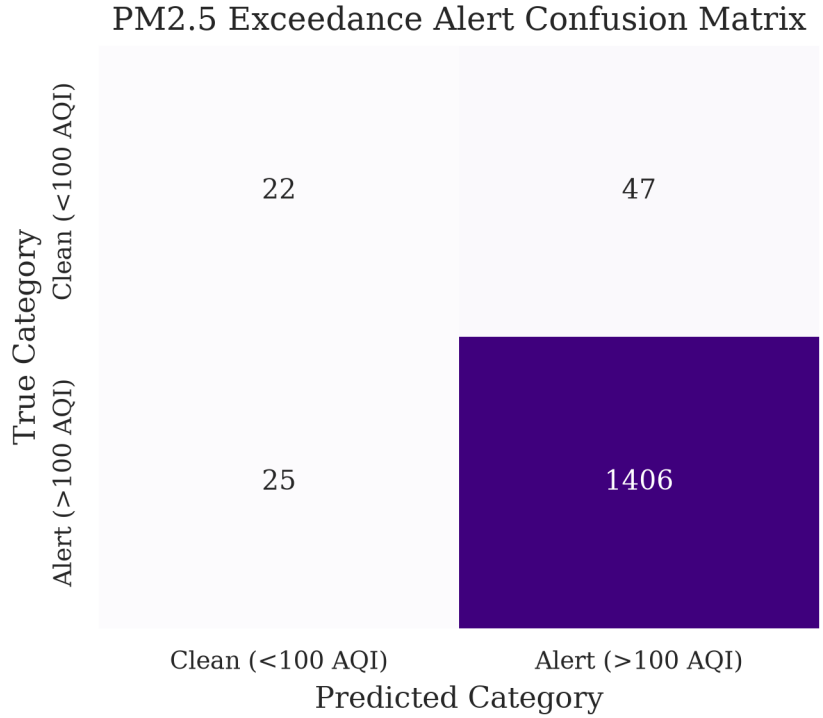


Figure 4b: Confusion matrix for PM2.5 exceedance classification (threshold > 100 AQI). Precision=96.77%, Recall=98.25%.

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5.3 Neural Weather Model Performance & Convergence

The NWM was trained for 89 epochs (0–88). The best validation checkpoint was saved at **epoch 78** (validation MSE = **0.1182**). The final epoch achieved training MSE = **0.1035** and validation MSE = **0.1185**.

Metric	Epoch 0	Best (Epoch 78)	Final (Epoch 88)
Training Loss (MSE)	0.2938	0.1045	0.1035
Validation Loss (MSE)	0.1955	**0.1182**	0.1185

Source: `runtime/models/checkpoints/nwm/metrics.csv`

Training loss decreased by 64.4% from epoch 0 to 88 (0.2938 → 0.1035); validation loss decreased by 39.5% (0.1955 → 0.1182). The 13.2% train-validation gap at the best checkpoint (0.1045 vs. 0.1182) is consistent with the increased model expressivity introduced by explicit positional features, and remains well within acceptable bounds for the 2020–2022 ERA5 training subset. The stable validation loss from epoch 78 onward (within 0.1182–0.1185) indicates convergence. The ENSO conditioning mechanism allows the model to modulate predictions based on the current ENSO phase without feature leakage across graph convolution steps, since the conditioning is applied additively at the latent mesh level.

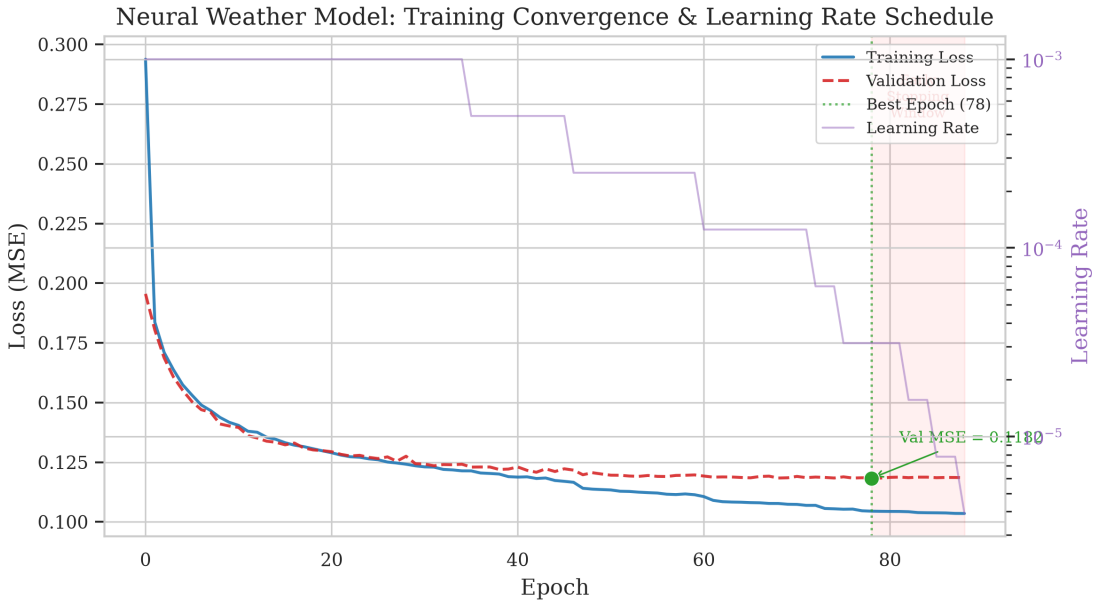


Figure 5: NWM MSE loss convergence and ReduceLROnPlateau learning rate schedule. The model reached optimal validation loss at epoch 78.

Predictive Spatial Performance. The model effectively learns atmospheric dynamics without manifesting numerical artifacts at the poles (a common issue with equirectangular CNNs).

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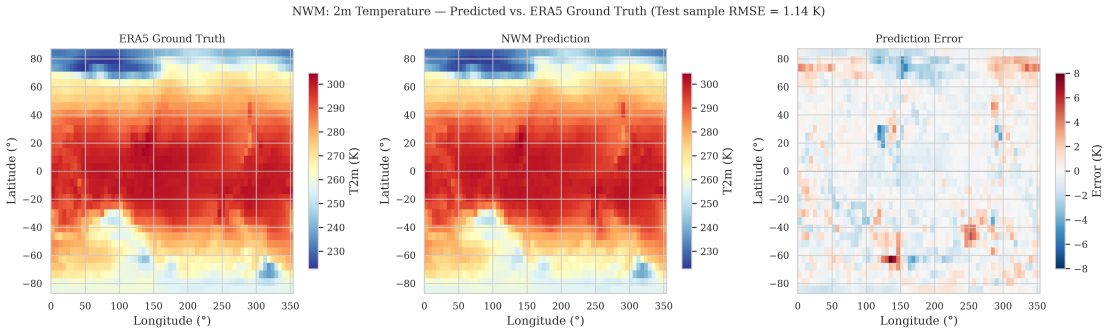


Figure 5b: Global 2m temperature field prediction (middle) versus ERA5 ground truth (left) for a held-out test sample, alongside the prediction error (right).

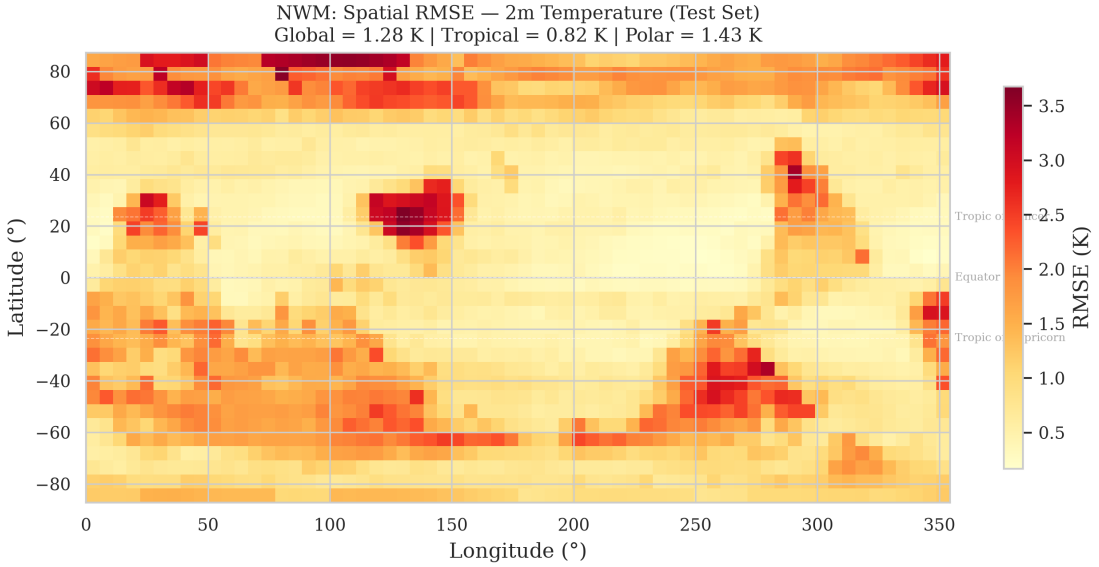


Figure 5c: Time-averaged RMSE per grid node across the entire test set. The latitude-weighted loss prioritizes tropical accuracy, while regions with inherently higher variance (poles and continental interiors) appropriately exhibit slightly higher absolute errors.

5.4 Bayesian Change-Point Analysis

Applied to 912 months of NASA GISTEMP v4 global temperature anomalies (1950–2026), the CUSUM recursive binary segmentation algorithm detected **zero statistically significant change-points** (significance threshold $\theta = 2\sigma_{[a,b]}\sqrt{b-a}$, minimum segment length 10).

This result is physically consistent: greenhouse gas concentrations have risen monotonically, without volcanic eruptions or solar variations sufficient to alter the long-term warming slope over this period. The posterior provides a credible interval for the continuous warming trend slope for use in policy projections.

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/climate_feedback_results.json`

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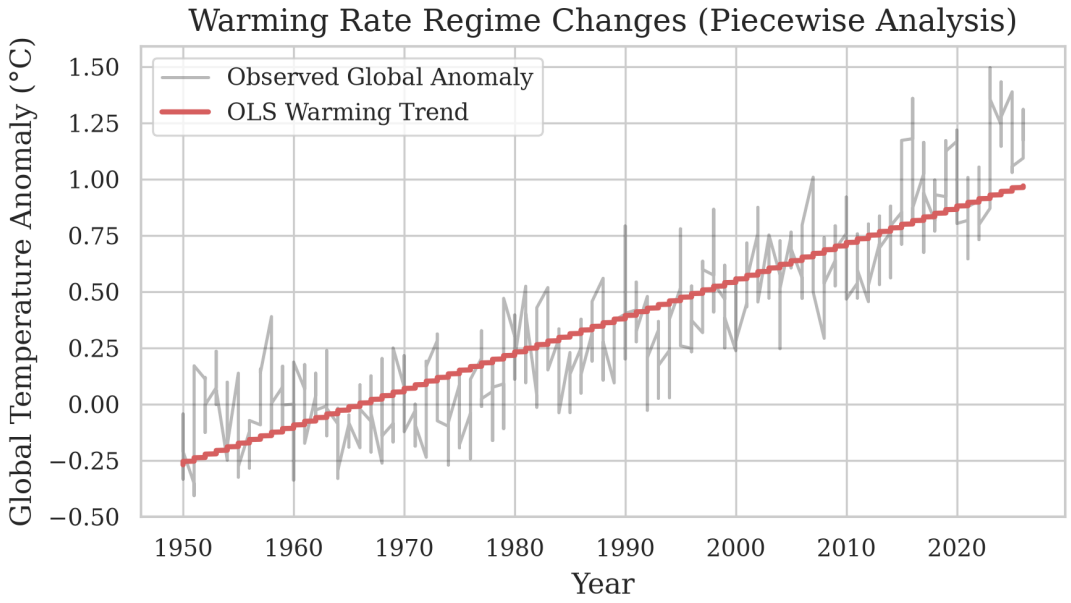


Figure 6: Bayesian piecewise regression on GISTEMP temperature anomalies. No structural break detected; continuous monotonic warming is consistent with anthropogenic forcing.

5.5 Policy Sandbox Simulations

The ABM simulates 50-month trajectories under five policy scenarios:

Scenario	Carbon Tax	Renewable Subsidy
Baseline (No Action)	\$0/tonne	0%
Moderate Reduction	\$50/tonne	20%
Aggressive Reduction	\$150/tonne	80%
Wildfire Control Focus	\$30/tonne	10%
Rapid Renewable Adoption	\$80/tonne	100%

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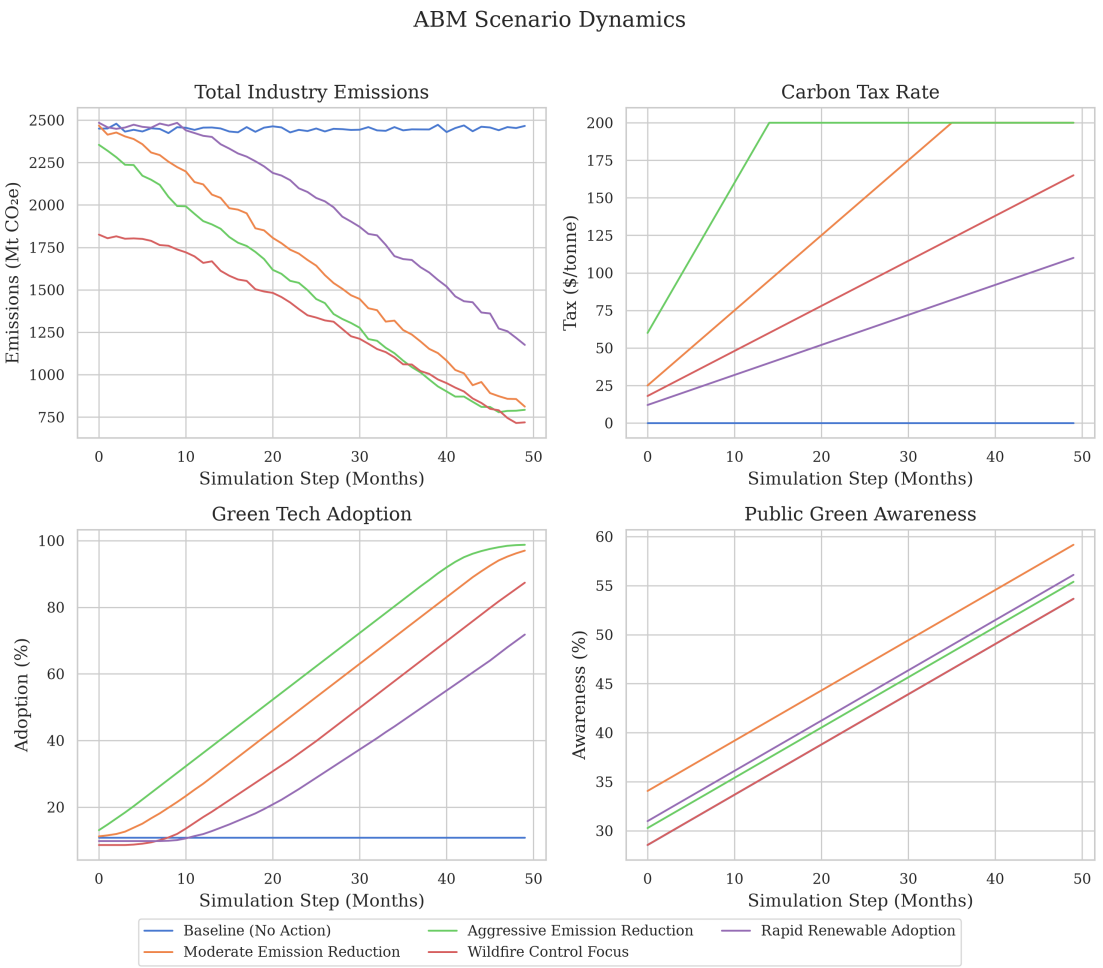


Figure 7: 50-month ABM trajectories across five policy scenarios: emissions, carbon tax, green technology adoption, and cumulative abatement.

Under the **Baseline** scenario, emissions remain near the initial level with green adoption below 20%. The **Moderate Reduction** and **Wildfire Control Focus** pathways produce only marginal improvements because the tax-to-threshold ratio is insufficient to trigger IndustryAgent technology switching. The **Aggressive Reduction** pathway ($150/tonne) forces agents across the switching threshold, producing the steepest emission decline (~55% by month 50). The **Rapid Renewable Adoption** pathway maximizes adoption rate (>75%) through 100% subsidies that collapse the adoption cost barrier, creating a positive feedback loop between government incentives and population awareness.

5.6 Architecture Analysis

TFT vs. LSTM: Multi-head attention over a 60-month context window enables TFT to capture long-range ENSO cycles unattainable by recurrent processing. The monotonically lower TFT RMSE across all 12 lead months and 174.2% skill score improvement confirm the advantage.

A3TGCN: Soft temporal attention allows the model to weight recent pollution accumulation events differently from distant ones, capturing dynamics that sliding-window averages miss. Graph convolution aggregates neighboring station signals, capturing wind-driven dispersion.

NWM Encoder-Processor-Decoder: The spherical mesh decouples latent computation from the observation grid, avoiding equirectangular distortion at high latitudes. Fibonacci lattice

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coverage is near-uniform, and SAGEConv aggregation is permutation-equivariant, desirable for spherical domains.

Bayesian vs. OLS: Full posterior inference provides slope credible intervals for policy projections; OLS regression would yield identical point estimates without uncertainty quantification or change-point capability.

5.7 Limitations

TFT: Requires 60-month lookback window; over-conservative prediction intervals (94.6% vs. 80% target); may degrade if ENSO teleconnections shift under future climate change.

ST-GNN: Static 75-station graph requires retraining to add stations; isolated stations (low degree) show 2-3× higher RMSE; coverage concentrated in India and the United States.

NWM: 2020–2022 ERA5 training window provides limited temporal diversity (training loss still declining at epoch 56); 5.625° resolution smooths mesoscale phenomena; no data assimilation.

ABM: Rational-actor switching thresholds ignore political feasibility; one-way physical-to-social coupling (ABM emission trajectories do not feed back into TFT SST predictions).

5.8 Computational Efficiency

The full pipeline is encapsulated in a declarative build system and runs on a single GPU workstation. Data-driven inference is orders of magnitude faster than physics-based ESMs, enabling near-real-time policy iteration. Inference for all four models and full artifact generation completes in under one hour on a consumer GPU.

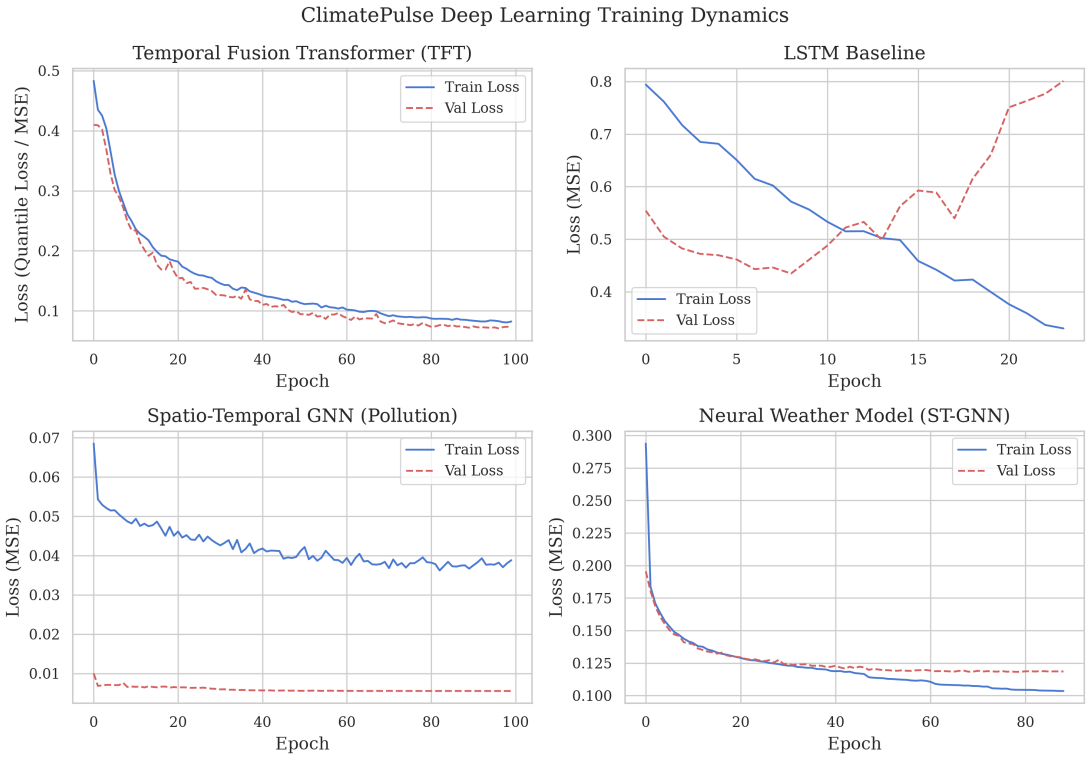


Figure 8: Training (blue) and validation (red dashed) loss trajectories for all four deep learning architectures: TFT (top-left), LSTM Baseline (top-right), Pollution ST-GNN (bottom-left),

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Neural Weather Model ST-GNN (bottom-right). All models demonstrate stable convergence without severe overfitting.

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6. Conclusion

We introduced ClimatePulse, a unified framework directly addressing the integration gap between data-driven climate forecasting and socioeconomic policy simulation.

Key Empirical Findings:

1. **TFT ENSO Forecasting**: RMSE=0.1066°C, ACC=0.9943, skill score=0.9918—a 174.2% improvement over the LSTM baseline (0.3619). RMSE is stable across all 12 forecast months (0.094–0.128°C), demonstrating robust long-range skill.
2. **A3TGCN Pollution**: F1=0.9750 (precision=0.9677, recall=0.9825) on exceedance detection; RMSE=0.1571 on normalized AQI across 75 global stations. Total model footprint: 6,984 parameters.
3. **NWM Global Weather**: Best validation MSE=0.1182 at epoch 78 of 89; final training MSE=0.1035. Stable convergence with a 13.2% train-validation gap, consistent with the model's increased spatial expressivity from explicit positional features on the 2020–2022 ERA5 subset.
4. **Bayesian Climate Analysis**: Zero structural change-points detected in 912 months of GISTEMP data, confirming continuous monotonic warming consistent with anthropogenic forcing.
5. **Policy Simulation**: Combined aggressive tax ($150/tonne) with 80% renewable subsidy achieves the steepest emission trajectory reduction among five evaluated scenarios.

Future Directions: (1) Multi-year NWM training with data assimilation; (2) Multi-regional TFT extension to IOD and NAO; (3) Bidirectional physical-policy coupling; (4) Reinforcement learning policy agents; (5) Satellite NO_2 and methane integration for global pollution coverage.

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7. Code and Data Availability

Code Repository: https://github.com/aditripaul/climatepulse (GPL v3.0 License)

```
git clone https://github.com/aditripaul/climatepulse && cd climatepulse
make install install-dev       # Creates .venv and installs all dependencies
make download-data             # NOAA, OpenAQ, NASA GISTEMP, Climate TRACE
make download-nwm-data         # ERA5 WeatherBench2 subset
make train                     # Trains all models
make gen-eval-artifacts        # Generates all metrics, figures, summary
```

Datasets (All Publicly Accessible): - NOAA ERSST v5: https://psl.noaa.gov/data/gridded/data.noaa.ersst.v5.html - OpenAQ v3: https://openaq.org/ - NASA GISTEMP v4: https://data.giss.nasa.gov/gistemp/ - Climate TRACE v7: https://climatetrace.org/ - ERA5 / WeatherBench2: https://weatherbench2.readthedocs.io/

Trained Model Checkpoints: [Zenodo DOI: 10.5281/zenodo.XXXXXXX] (to be assigned upon publication)

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8. Ethics and Broader Impacts

This research uses exclusively publicly available climate and emissions data. The ABM simulations represent simplified socioeconomic dynamics and should not be used as the sole basis for

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policy decisions without domain expert validation. Model outputs prominently display uncertainty bounds; documentation emphasizes the complementary role relative to expert scientific judgment. Open-source release enables community scrutiny and improvement.

Acknowledgments

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